# Kai Wu

Department of Economics, The University of Texas at Austin
2225 Speedway, BRB 3.150, Austin, TX 78712
kaiwu@utexas.edu | +1 (512) 200-5572 | kaiwu-econ.com

## Education

**The University of Texas at Austin**, Ph.D. in Economics — Aug. 2021 – Present
**Jinan University**, M.A. in Economics — Sept. 2018 – Jun. 2021
**Northwest University** (China), B.A. in Economics — Sept. 2014 – Jun. 2018

## Research Fields

**Primary:** Public Economics, Networks/International Trade, Empirical Industrial Organization
**Secondary:** Urban Economics, Environmental Economics, Political Economy

## References

**Marika Cabral** (Primary)
Department of Economics
UT Austin
marika.cabral@utexas.edu

**Nitya Pandalai-Nayar**
Department of Economics
UT Austin
npnayar@utexas.edu

**Jackson Dorsey**
Department of Economics
UT Austin
jackson.dorsey@austin.utexas.edu

**Cody Tuttle**
Department of Economics
UT Austin
cody.tuttle@utexas.edu

**Ben Zou**
Department of Economics
Purdue University
zou136@purdue.edu

## Working Papers

**Crowding** (with Yizhen Gu, Qu Tang, and Ben Zou) [Job Market Paper]

**Abstract:** Crowding is a common disamenity, yet little is known about its monetary cost. We estimate the willingness to pay (WTP) to avoid crowding in public transit using administrative smartcard trip records from the Beijing Subway and a revealed-preference approach. Passengers choose departure times, trading off fares, in-train crowding, and schedule deviation, with fare variation from an early-bird discount. Leveraging the subway's network structure, we impute real-time in-train crowding, infer each passenger's optimal arrival time, and construct a novel instrumental variable for crowding. The marginal WTP to reduce crowding by one passenger per square meter is about 0.05 RMB per minute, implying a crowding externality comparable to the fare. High-income passengers have higher WTP to avoid crowding but are less price-sensitive. An optimal crowding tax improves welfare within the subway system but is regressive; a two-class configuration elicits self-selection and benefits both high- and low-income commuters.

**Beyond Targeted Firms: Supply Chain Spillovers of Environmental Regulation in China** (with Xinghua Deng) [*Draft coming soon*]

**Abstract:** This paper studies the cost propagation of environmental regulation along production networks. We exploit a natural experiment, the 2017 Xiong'an pollution shutdown campaign in China, which imposed mandatory closures on polluting industrial operations across a newly designated development zone, using administrative firm-to-firm transaction data. The targeted firms' purchases and sales fell sharply, and the shock propagated in both directions: upstream suppliers lost sales and downstream customers cut purchases. Losses further extend to higher order trade partners. The effects are driven by two-sided complementarity: when a firm loses a trading partner, it also contracts trade with its remaining partners, both its purchases from unaffected suppliers (input side) and its sales to unaffected customers (output side). A general equilibrium model of production

networks with scale economies formalizes these patterns and shows that most of the total GDP loss falls outside the targeted area. Ignoring these network spillovers leads regulators to overregulate, requiring more abatement than is socially optimal. A single emissions tax addresses both environmental and network externalities at once by sorting firms on private abatement cost and network connectivity.

### Political Consolidation and Corporate Tax Burden

**Abstract:** I investigate the impact of political consolidation on the corporate tax burden, leveraging the variation from a nationwide reform in China that shifted governance autonomy from annexed to annexing political units. Using administrative firm-level tax payment data from 2008 to 2015 and a heterogeneous difference-in-differences approach, I find divergent shifts in the effective corporate income tax rate among affected areas following the annexation. On average, annexing areas experienced a 1.8% decrease in the effective tax rate, contrasted with a 1.4% increase in annexed areas. These changes worked to make corporate tax rates more uniform across annexed and annexing areas. The observed effects are largely attributed to the adjustment in less transparent tax break programs, over which local governments possess greater discretion. Heterogeneity analysis suggests political favoritism and tax competition are additional mechanisms contributing to these findings.

## Selected Work in Progress

**Stock Market Turbulence and Production Networks** (with Husang Kim and Rikuto Onishi)

We study whether non-fundamental stock price volatility propagates through production networks, exploiting the 2015 Chinese stock market crash, a liquidity crisis driven by retail panic rather than firm fundamentals. Using cross-sectional variation in idiosyncratic volatility orthogonal to market and industry movements, we find that firms with greater price disruption experienced significant declines in sales and purchases, with evidence of propagation to trading partners.

## Conference and Seminar Presentations

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| (* indicates scheduled) UT Austin; ABFER Annual Conference; SEA Annual Meeting* (nominated student) | 2026 |
| Texas Applied Microeconomics Student Workshop | 2025 |

## Fellowships and Awards

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| Summer Excellence Fellowship, Graduate School, The University of Texas at Austin | 2026 |

## Teaching Experience

*Teaching Assistant, The University of Texas at Austin*

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| Introduction to Microeconomics | 2021, 2022, 2024, 2026 |
| Urban Economics | 2022 |
| Business Strategy | 2023 |
| Industrial Organization | 2023 |
| Economics of Education | 2025 |
| Energy and Environmental Economics | 2025 |

## Research Experience

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| Research Assistant for Prof. Yizhen Gu | 2018–2021 |
| Research Assistant for Prof. Marika Cabral | 2023–2025 |

## Skills

**Software:** Stata, Python, R, SQL, ArcGIS, QGIS, LaTeX
**Languages:** English (fluent), Chinese (native)